

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
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Experiment 7
Bagging, Boosting, and Stacked Ensemble Models

1. Objective

- To understand ensemble learning strategies: Bagging, Boosting, and Stacking.
- To implement Bagging and Boosting classifiers.
- To build a Stacked Ensemble model using multiple base learners.
- To compare ensemble models in terms of accuracy, stability, and generalization.
- To analyze the effect of ensemble methods on bias and variance.

2. Dataset

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant (M) and Benign (B)

3. Theory

Bagging (Bootstrap Aggregation)

Bagging is an ensemble technique that trains multiple models on different bootstrap samples of the training data and aggregates their predictions.

Key Characteristics:

- Reduces variance

- Suitable for high-variance models
- Independent model training

Boosting

Boosting is an ensemble method where models are trained sequentially, and each new model focuses on correcting the errors made by previous models.

Key Characteristics:

- Reduces bias
- Sequential learning
- Emphasizes difficult samples

Common boosting algorithms include AdaBoost and Gradient Boosting.

Stacked Ensemble (Stacking)

Stacking combines multiple heterogeneous base models and trains a meta-learner to optimally combine their predictions.

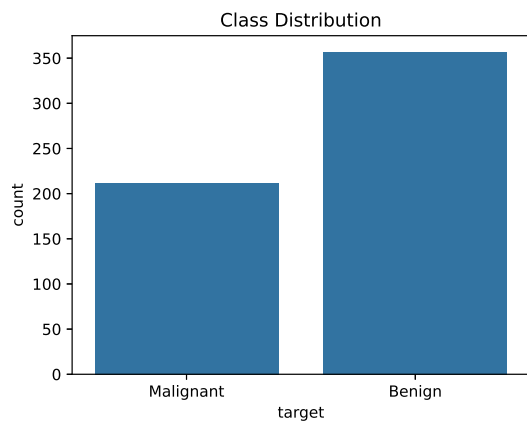
Key Characteristics:

- Uses diverse base learners
- Meta-model learns optimal combination
- Often achieves superior performance

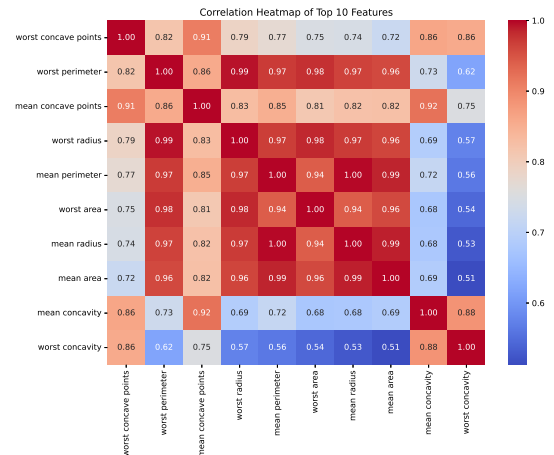
4. Implementation Details

- Loaded and preprocessed the dataset.
- Performed exploratory data analysis (EDA).
- Split the dataset into training and testing sets using an 80–20 ratio.
- Implemented a Bagging classifier using a Decision Tree base estimator.
- Implemented Boosting classifiers such as AdaBoost and Gradient Boosting.
- Constructed a Stacked Ensemble using multiple base models.
- Defined hyperparameter search spaces for each ensemble model.
- Selected optimal hyperparameters using 5-Fold Cross-Validation.
- Evaluated all models using standard performance metrics.
- Compared results and analyzed bias–variance behavior.

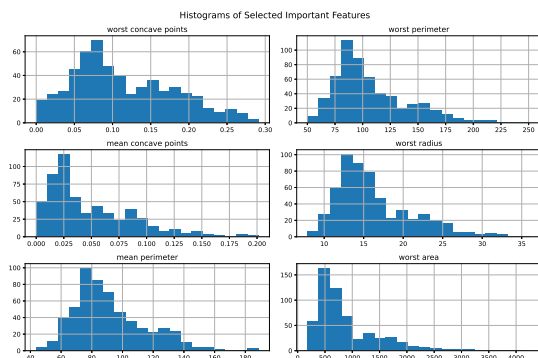
5. Visualizations



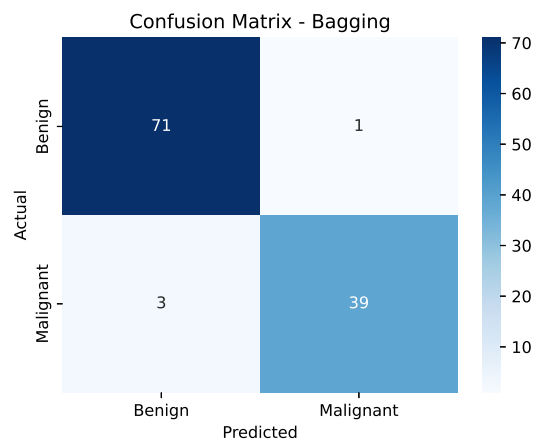
(a) Correlation heatmap



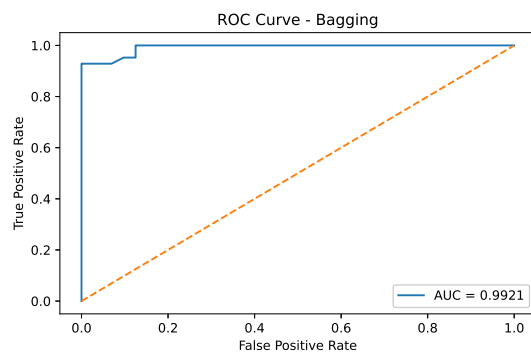
(b) Feature distribution



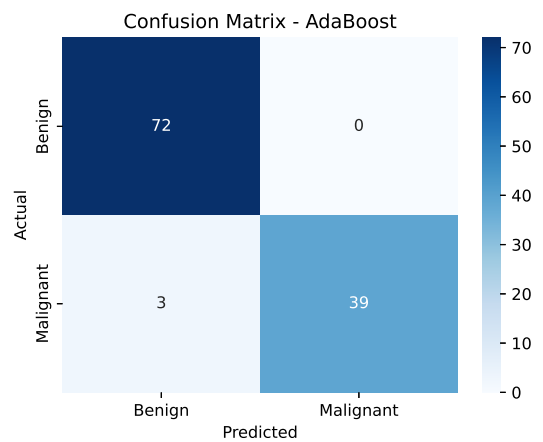
(a) Feature Histogram



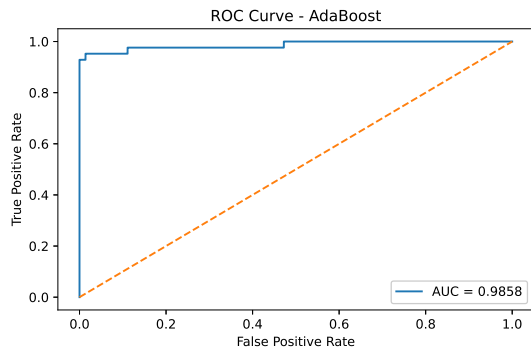
(b) Bagging Confusion Matrix



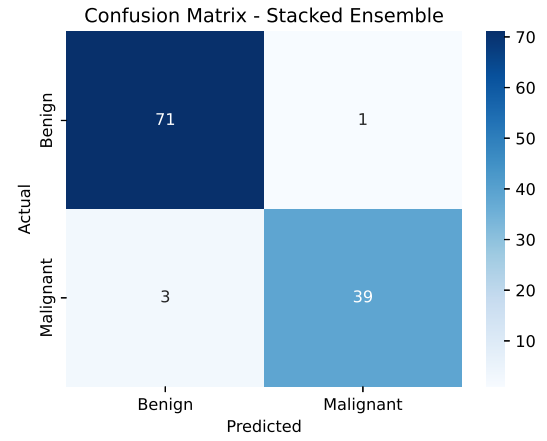
(a) Roc Bagging



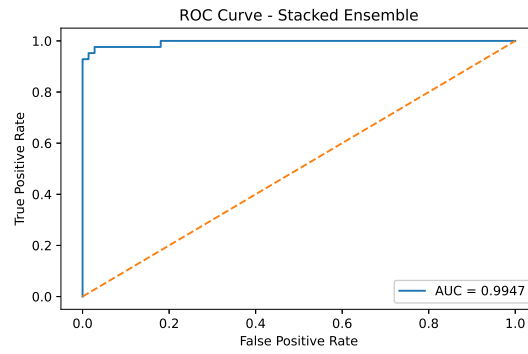
(b) Boosting Confusion Matrix



(a) Roc Boosting



(b) Ensemble Stacking Confusion Matrix



(a) ROC Stacking

6. Hyperparameter Tuning Results

Bagging Results

Table 1: Bagging Hyperparameter Evaluation

n_estimators	max_samples	Avg CV Accuracy (%)	Avg CV F1 Score
50	0.7	96.48	0.9515

Boosting Results

Table 2: Boosting Hyperparameter Evaluation

n_estimators	learning_rate	Avg CV Accuracy (%)	Avg CV F1 Score
100	0.50	97.14	0.9607

Stacked Ensemble Results

Table 3: Stacked Ensemble Evaluation

Base Models	Meta Learner	Avg CV Accuracy (%)	Avg CV F1 Score
SVM + Naive Bayes + Decision Tree	Logistic Regression	96.92	0.95

7. Performance Comparison

Table 4: Performance Comparison of Ensemble Models

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	96.49	0.975	0.9286	0.9512
Boosting	97.37	1.000	0.9286	0.9630
Stacked Ensemble	96.49	0.975	0.9286	0.9512

8. Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC Curve and AUC

9. Observation Questions and Answers

1. How does Bagging reduce variance?

Bagging reduces variance by training multiple decision trees on different bootstrap samples and averaging their predictions. Because each tree sees a slightly different dataset, the model becomes less sensitive to fluctuations in any one training sample.

2. How does Boosting address model bias?

Boosting builds models sequentially, where each new learner focuses more on the samples that were misclassified earlier. This step-by-step correction helps the ensemble learn patterns that a weak learner may initially miss, thereby reducing bias.

3. Why does stacking benefit from heterogeneous models?

Stacking combines different kinds of base learners such as SVM, Naive Bayes, and Decision Tree. Since these models make different types of errors, the meta-learner can combine their strengths and produce a stronger final prediction.

4. Which ensemble method performed best and why?

The best model on the test set was: AdaBoost with accuracy = 97.37%. It performed best because its ensemble design produced the strongest balance between accuracy, recall, F1-score, and AUC on this dataset.

10. Conclusion

implemented Bagging, Boosting, and Stacked Ensemble models on the Wisconsin Diagnostic Breast Cancer dataset. Using 5-fold cross-validation, the models were tuned and compared using accuracy, precision, recall, F1-score, confusion matrix, ROC curve, and AUC.